Lab Tasks

Task # 01:

You are asked to create a Java class named Box with two attributes, width and height, and a method print() that displays these values, following the structure shown in the provided sample code. Once you have implemented the basic class with these attributes and method, your next task is to modify it by adding constructors with different ways of initializing a Box object:

1. A constructor that takes both width and height as parameters and only creates the object if both values are positive.
2. A constructor that takes only height as a parameter and prompts the user to input the width.
3. A constructor with no parameters that asks the user to input both width and height.

This exercise will help you practice creating multiple constructors, validating input, and interacting with the user while reinforcing your understanding of class design in Java.

Task # 02:

A bookstore requires a program to manage its book inventory. Design a Java class named Book with three instance variables: title (String) to store the book's title, author (String) to store the author's name, and price (double) to store the book's price. Implement a parameterized constructor to initialize all these fields when a Book object is created. Provide setter methods to allow updating the title, author, and price, and getter methods to retrieve these details. The class should ensure controlled access to the book information and allow accurate management of book data through these methods.

Task # 03:

You are tasked with creating a Java class that represents a programming language. The class should have a String variable named language and include two constructors: a default constructor that sets language to "JAVA" and a parameterized constructor that allows the value of language to be set when the object is created. Additionally, the class should have a method that prints the value of

language. In the main method, you are required to create two objects of this class: one using the default constructor and one using the parameterized constructor with the value "C", and then display the languages stored in both objects. This exercise will help you practice working with constructors, class variables, methods, and object creation in Java.

Task # 04:

Create a class called Holiday to represent holidays with three instance variables: name (a string for the holiday's name), day (an integer for the day of the month), and month (a string for the month). The class should include a constructor that initializes these three variables using values provided when a new Holiday object is created. Additionally, implement a method called inSameMonth that takes another Holiday object as input and returns true if both holidays occur in the same month, and false otherwise. Finally, demonstrate the class by creating a Holiday instance named "Independence Day" that occurs on the 14th of August.

Task # 04(b):

Implement a copy constructor for Holiday class and demonstrate its use in the main function.

Task # 05:

A computer store needs a program to manage its laptop inventory efficiently. Design a Java class named Laptop with the following private instance variables: brand (String), model (String), price (double), and stock (int). Implement constructor overloading by providing a default constructor, a parameterized constructor that initializes all fields using the this keyword, and a copy constructor that creates a new Laptop object by copying another object's data. Include setter and getter methods to update and retrieve each field, demonstrating encapsulation. At runtime, create Laptop objects dynamically and use the copy constructor to duplicate an existing laptop. Implement a method to indicate when a laptop object is being removed, simulating a destructor. Finally, display the details of all laptops, ensuring accurate stock and price management throughout.

Task # 06:

A bookstore requires a program to manage its book inventory. Design a Java class named Book with three instance variables: title (String) to store the book's title, author (String) to store the author's name, and price (double) to store the book's price. Implement a parameterized constructor to initialize all these fields when a Book object is created. Provide setter methods to allow updating the title, author, and price, and getter methods to retrieve these details. The class should ensure controlled access to the book information and allow accurate management of book data through these methods.

Task # 07:

A company wants a program to manage employee records and calculate monthly payroll. Design a Java class Employee with private variables: empID (int), name (String), basicSalary (double), and daysWorked (int). Implement constructor overloading with a default constructor, a parameterized constructor (using this), and a copy constructor. Provide setter and getter methods for all variables (encapsulation) and create objects dynamically at runtime. Include a method calculateSalary() to compute (basicSalary / 30) * daysWorked and a destructor/finalizer to indicate object removal.

Tasks: Create at least three employees using different constructors, update their details, duplicate a record using the copy constructor, and display all employee details along with calculated salaries.

Task # 08:

Create a class named as "GarbageCollection":

- Contains a constructor that prints "Object created".
- A finalize( ) method that prints "Object deleted".

Perform object deletion using the following methods:

1. Null a reference: Create an object then set the reference to null.
2. By assigning a reference to another object using assignment operator.
3. By creating an anonymous object.